Fiscal Year:	FY 2013	**Task Last Updated:**	FY 04/08/2013
PI Name:	Gibble, Kurt Ph.D.		
Project Title:	Contributions to the Evaluation of the ACES Clock PHARAO		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	16802-6300	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Fundamental Physics NNH11ZTT001N
Start Date:	06/08/2012	**End Date:**	06/08/2017
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	JPL 1459238		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	ACES (Atomic Clock Ensemble in Space) is an ESA project that will test fundamental physics, search for the time variation of fundamental constants, and enable highly precise frequency comparisons between clocks on Earth. It is planned to operate for 18 months on the International Space Station. PHARAO (Projet d'Horloge Atomique par Refroidissement d'Atomes en Orbit) is the laser-cooled, microgravity, atomic cesium clock of ACES that provides state-of-the-art accuracy, with a goal of 1 part in 10^(16) accuracy. We propose to assist in the accuracy evaluation of PHARAO, a central scientific objective of the ESA mission. We will contribute to the analysis of the three largest sources of systematic uncertainty: clock frequency shifts due to the collisions of ultracold atoms, a Doppler shift known as the distributed cavity phase shift, and a frequency shift due to the lensing of the atomic wavepackets by the clock's microwave field. The current working uncertainty budget projects 2.2×10^(-16) accuracy. The proposed work will reduce the largest contribution to the clock's uncertainty to negligible levels and investigate significant reductions of the others, with a goal of reaching 10^(-16) accuracy. The work will help define the limits of PHARAO's accuracy and identify		

	workable procedures to demonstrate its accuracy.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our cavity work produced an improved microwave cavity design that will be used for Cs and Rb fountain clocks. These are being incorporated into new clocks being built at the British and South Korean National Laboratories, National Physical Laboratory (NPL), and KRISS. The new cavity design allows significantly reduced clock errors due to Doppler shifts.
Task Progress:	We have constructed preliminary 3D Finite Element Models of the electromagnetic fields in the PHARAO clock and state-selection microwave cavities. The models have been scaled up to use our full memory of 1TB. The solutions were used to calculate distributed cavity phase errors. The results showed the different behaviors for PHARAO on the ground, in space, and as compared to fountain clocks. They also show that the detuned selection technique that we proposed reduces the variation of the state selection probability by more than a factor of 30. This will reduce all of the largest systematic errors for PHARAO, ultracold collisions, distributed cavity phase, and microwave lensing frequency shifts. More work is required to establish uncertainties due to details of the cavity geometry and construction, and RAM limitations. We completed designs of a microwave cavities for several new primary fountain clocks based on cesium and rubidium. Our designs have lower cavity phase errors than existing cavities. Several national laboratories around the world have built these cavities and are in the process of building the atomic clocks. On our measurements of Cs Feshbach resonances, we have completed a number of optimizations and have made precise measurements of scattering phase shifts through Feshbach resonances. We have evaluated and reduced a number of systematic errors in the measurements. Evaluating systematic errors is ongoing, before averaging to yield high precision. Talks were presented on the accuracy evaluation and models of the PHARAO Microwave cavity at the ASGSR meeting in New Orleans and at the ACES Working Group meeting in Paris. A talk was presented at CPEM 2012 in Washington, DC showing the optimized design of a microwave clock cavity.
Bibliography Type:	Description: (Last Updated: 05/28/2026)
Abstracts for Journals and Proceedings	Gibble K, Peterman P, Moric I, Laurent P, Grosjean O, Salomon C. "Accuracy Evaluation of the ACES Clock PHARAO." 28th Annual Meeting of the American Society for Gravitational and Space Research, New Orleans, LA, November 27-December 2, 2012. 28th Annual Meeting of the American Society for Gravitational and Space Research, New Orleans, LA, November 27-December 2, 2012. Program and abstracts, p. 66. See https://www.asgsr.org/images/stories/pdf/2012_AnnualMeeting_Program.pdf , Nov-2012
Papers from Meeting Proceedings	Gibble K, Lea SN, Szymaniec K. "A microwave cavity designed to minimize distributed cavity phase errors in a primary cesium frequency standard." 2012 IEEE Conference on Precision Electromagnetic Measurements, Washington, D.C., July 1-6, 2012. 2012 IEEE Conference on Precision Electromagnetic Measurements (CPEM) 2012 Conference Digest, p. 700-701. http://dx.doi.org/10.1109/CPEM.2012.6251122 , Jul-2012